ADVANCED PROBLEMS FOR LECTURE 13

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

For these questions, you will need to know the Mean Value Theorem. We will introduce this in the second half of the lecture, but you can look at the previous problem set for the statement and idea of the proof if you like.

(1) Prove Cauchy's Mean Value Theorem, which states that if f and g are both continuous on $[a, b]$ and differentiable on (a, b) then there is some $c \in (a, b)$ such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$$

(2) See if you can use the Mean Value Theorem to give a more rigorous argument for a *weak* fundamental theorem of algebra which says that a polynomial of degree n can have no more than n distinct roots. (Hint: start with degree 1).

(3) Prove a fixed point theorem on $\mathbb{R}$, which says that if f is continuous and differentiable on a (connected) subset of $\mathbb{R}$, and $f'(x) \neq 1$ for all x, then there is at most one number a such that $f(a) = a$. What goes wrong if our domain is not connected?

(4) Prove that if a function has derivative 0 on a connected subset of $\mathbb{R}$ then it is constant.